

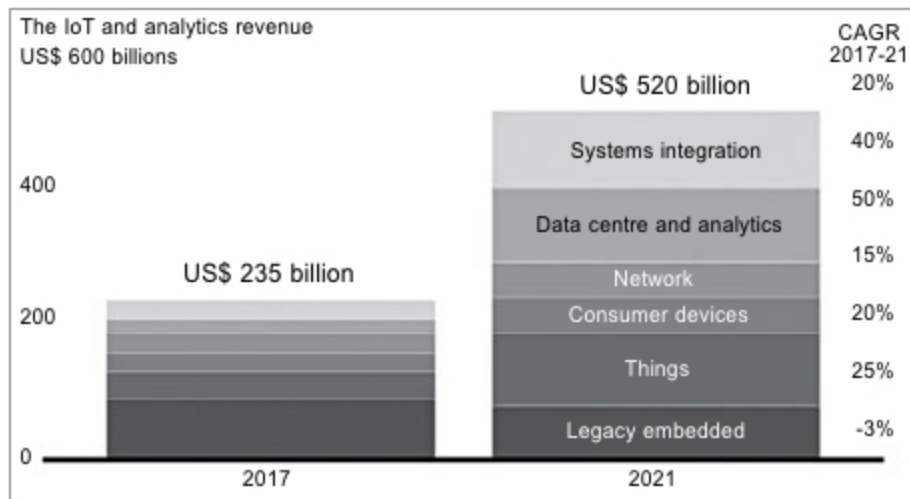


Fig. 1. Forecast on the growth of the global IoT market (Source: Bain Analysis)

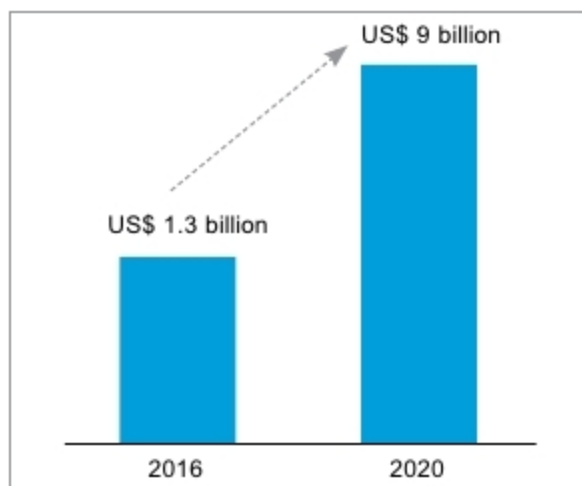


Fig. 2. The Indian IoT market (Source: Deloitte)

the combined markets of the IoT will grow to about US\$ 520 billion in 2021, more than double the US\$ 235 billion spent in 2017 (Fig. 1).

The Indian market scenario

India is rapidly growing as an IoT hub. Although, the country began its IoT journey much later than developed economies, its installed base of connected units is expected to grow at a rate much faster than them. According to a Deloitte report, the IoT units in India are expected to see a rapid growth of ~32x to reach 1.9 billion units by 2020, from its current base of 60 million. As a result, the Indian IoT market is expected to grow ~7x to move from US\$ 1.3 billion in 2016 to US\$ 9 billion by 2020 (Fig. 2). Key technology factors enabling IoT adoption in India include sensors, networks, standards, augmented intelligence and augmented behaviour (Table I).

Rise of the tech-savvy consumer along with increasing smartphone and mobile Internet penetration is driving consumer IoT applications in the Indian market. However, the consumer IoT adoption is expected

to be slower than the IoT adoption in industrial applications due to cost of IoT devices, security as well as privacy concerns of consumers.

The Indian IoT ecosystem consists of a diverse set of stakeholders including hardware vendors, software application vendors, network operators and system integrators. Interestingly, 60 to 65 per cent of these stakeholders are startups. The IoT presents opportunities for all stakeholders across the value chain.

However, according to the Deloitte study, application vendors focussing on both vertical and horizontal solutions, including consumer and industrial IoT (IIoT), are expected to garner 50 per cent share of the Indian IoT market. Along with applications suitable for a large segment of consumers, customised solutions for niche consumer groups are also gaining impor-

tance. Hardware vendors are focussing more and more on R&D to develop and expand their product portfolio in the IoT and to enhance their market share through mergers and acquisitions.

Network operators are increasing investments in new networks, such as SigFox, to increase connectivity revenues. System integrators are focussing majorly on IoT consultancy and implementation services, while enhancing digital capabilities and building solutions for the IoT through acquisition of niche companies and platforms. Start-ups with innovative solution portfolios are playing a significant role in driving the growth of the IoT in India.

Applications in demand

The IoT adoption in India is expected to grow across industries. Some key industry sectors driving it are transportation and logistics, utilities, manufacturing and automotive. These are expected to see highest adoption levels in India due to direct association with smart city projects. Moreover, sectors such as healthcare, retail and agriculture are also expected to make significant progress in IoT adoption.

Policy boost for IoT adoption in India

The government of India has a clearly-outlined IoT policy and related regulatory framework in place. The published IoT policy contains IoT-specific training provisions as well as R&D initiatives to support new IoT-enabled products and services. The government's planned investment worth US\$ 1 billion for 100 Smart Cities is a key enabler for IoT adoption.

In 2017, the World Bank cited the government's Digital India campaign as a key driver to making firms more competitive through the application of IoT technologies. An efficient IoT system in public administration translates into a better connected, secure and environment-friendly society.

TABLE I KEY TECHNOLOGY FACTORS ENABLING IoT ADOPTION IN INDIA		
Technology	Definition	Examples
Sensors	Device that generates an electronic signal from a physical condition or event	Small, robust and inexpensive sensors create information from conditions in the surroundings
Networks	Mechanism for communicating an electronic signal	Wireless networks with ubiquitous coverage such as cellular connectivity, Wi-Fi or LPWA networks
Standards	Commonly-accepted prohibitions or prescriptions for process framework	Technical standards enable processing of data and also interoperability among different devices
Augmented intelligence	Analytical tools that improve the ability to describe, predict and exploit relationships among phenomena	Databases with unstructured information are searched and analysed for corrective future actions
Augmented behaviour	Technologies and techniques that improve compliance with prescribed action	Several machine-to-machine interfaces remove fallible human intervention

Source: Deloitte